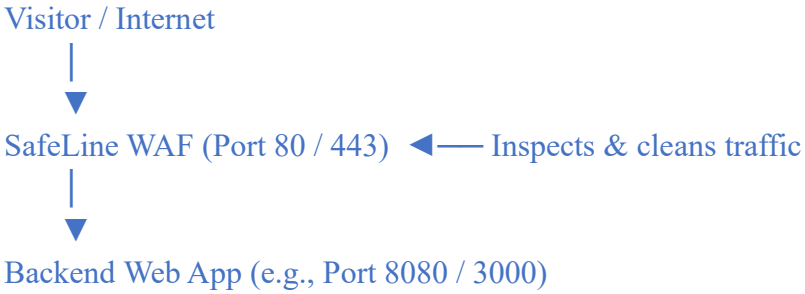
Hean Chhanun

SafeLine

1. What is SafeLine ?

SafeLine (by Chaitin Tech) is an open-source Web Application Firewall (WAF) and reverse proxy. It inspects incoming HTTP/HTTPS traffic using a semantic analysis engine to block web attacks (SQLi, XSS, code injection, brute force, bots, and Layer 7 DDoS) before they reach your backend application.



2. Prerequisites & System Requirements

- **Operating System :** Ubuntu 20.04 / 22.04 / 24.04 LTS (x86_64 or ARM64)
- **Hardware :** Minimum 2 CPU cores, 2 GB RAM, 10 GB free disk space
- **Network Port :**
  - 9443 (TCP) — SafeLine Management Web Console
  - 80 / 443 (TCP) — HTTP / HTTPS Traffic Inspection

3. How to install and setup SafeLine ?

1. **Step 1: Install Docker**

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y docker.io docker-compose-v2
sudo systemctl enable --now docker
```

2. **Step 2: Install SafeLine Waf**

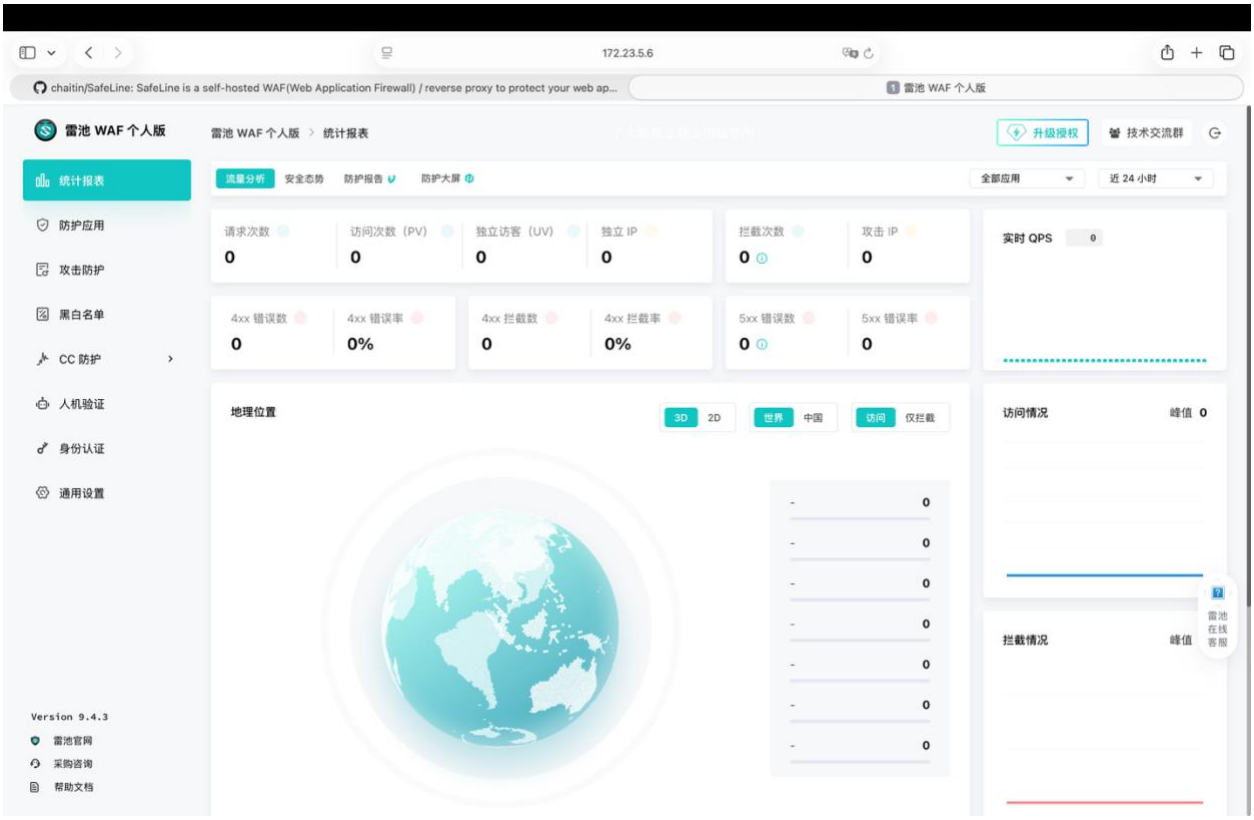
```
# 1. Run the official installer
bash -c "$(curl -fsSLk https://waf.chaitin.com/release/latest/setup.sh)"

# 2. (Optional) If download disconnects, resume and start containers:
cd /data/safeline
docker compose up -d
```

- When prompted for installation directory, press **Enter** (default: /data/safeline).
- When prompted for disk space confirmation, type **Y** and press **Enter**.

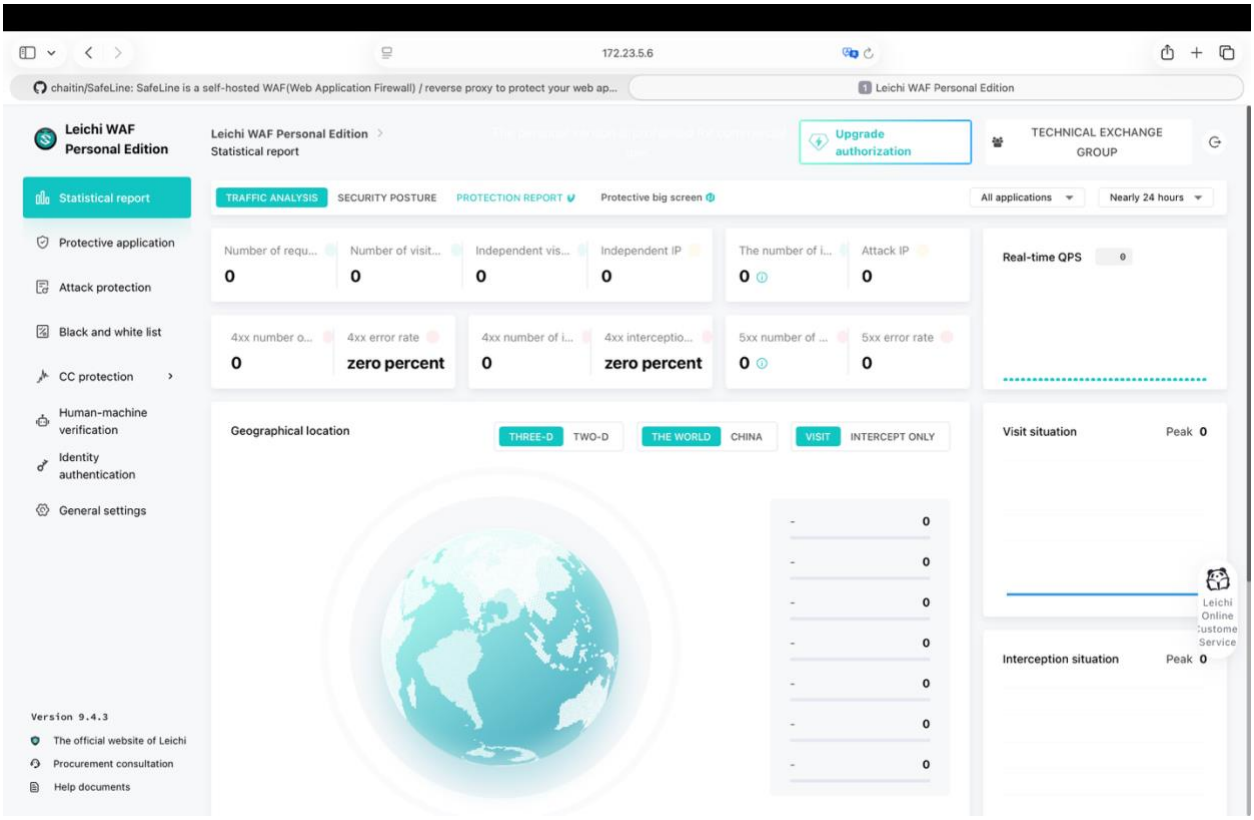
3. **Step 3: Set Custom Admin Password**
   SafeLine Default admin username is admin. Set desired password via terminal.
   docker exec -it safeline-mgt resetadmin -p "YourStrongPassword123!"

4. **Step 4: Open Management Console**
   1. Open browser: https://<YOUR-SERVER-IP>:9443
   2. Login with :
      - admin
      - YourStrongPassword123



5. **Step 5: Switch Interface to English**
   SafeLine Community Edition UI is natively in Chinese. To use it in English:
   - **Safari**: Click the Translation icon ([文A] / speech bubble) on the right side of the URL bar ➔ Click Translate to English.
   - **Chrome / Edge**: Right-click anywhere on the page ➔ Select Translate to English.

Hean Chhanun

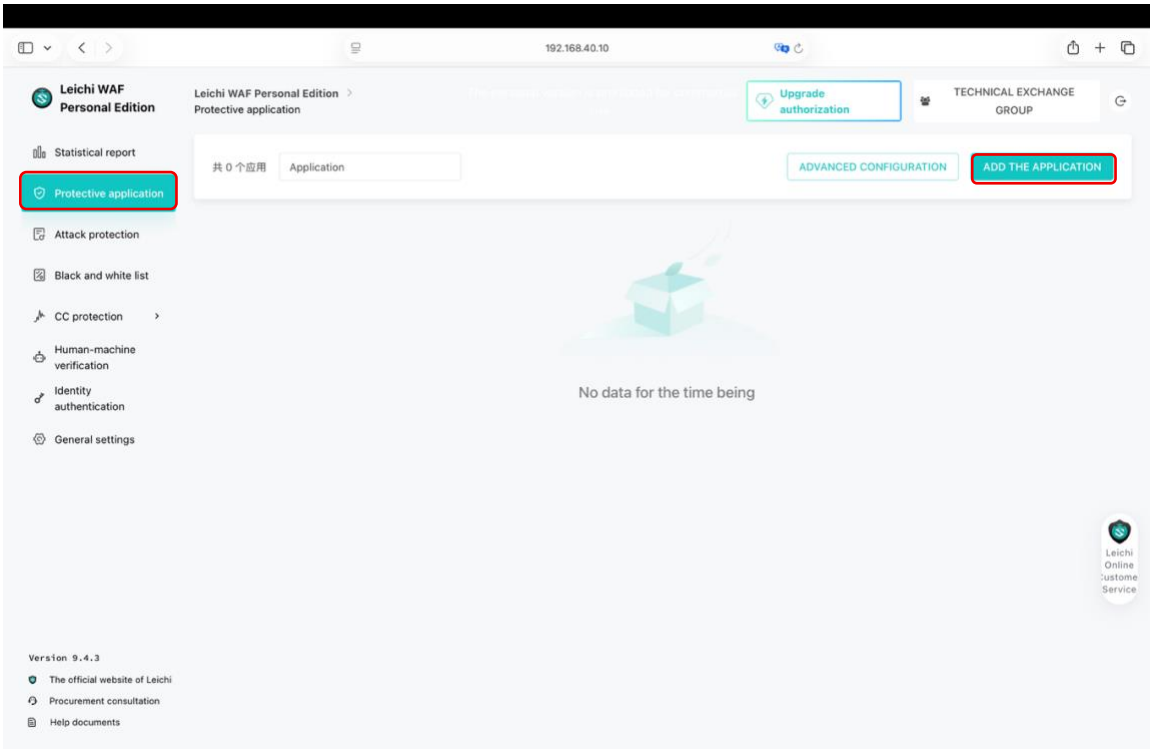


Note: Common Maintenance Commands

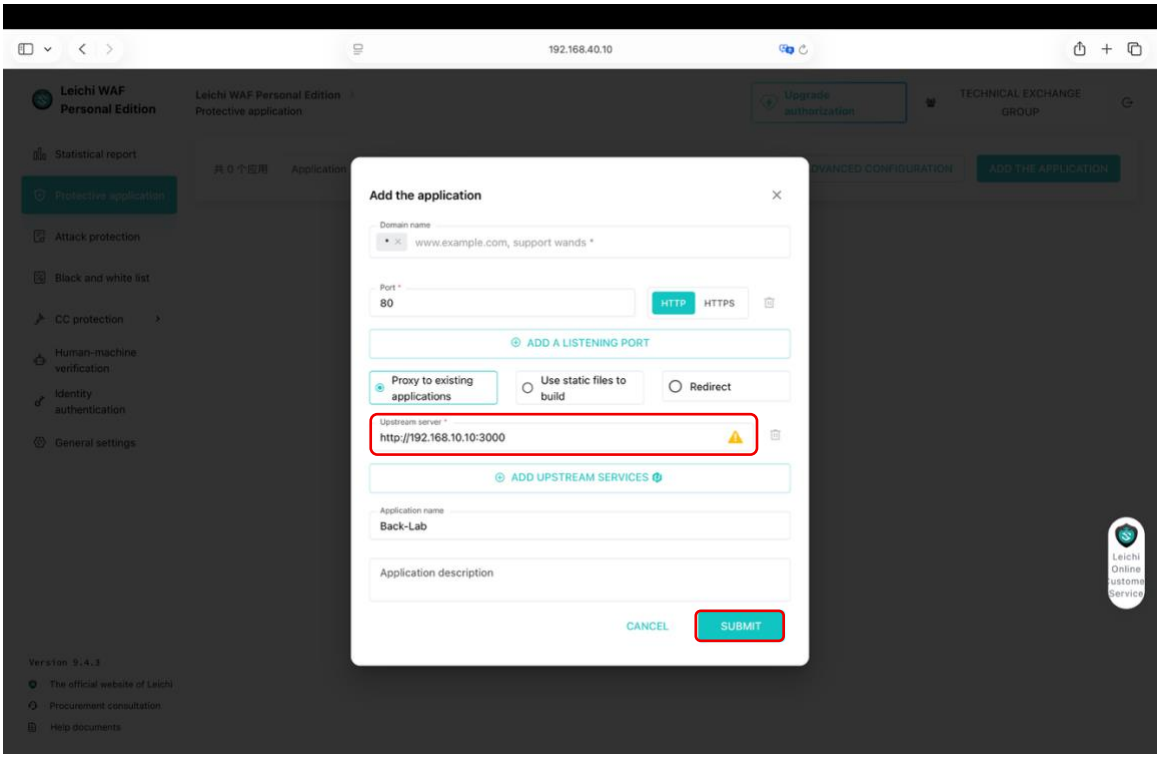
Action	Command
Check service status	`cd /data/safeline && docker compose ps`
View logs	`docker compose logs -f`
Restart SafeLine	`cd /data/safeline && docker compose restart`
Stop SafeLine	`cd /data/safeline && docker compose down`
Reset Admin Password	`docker exec -it safeline-mgt resetadmin -p "NewPassword"`

4. Apply Web with SafeLine

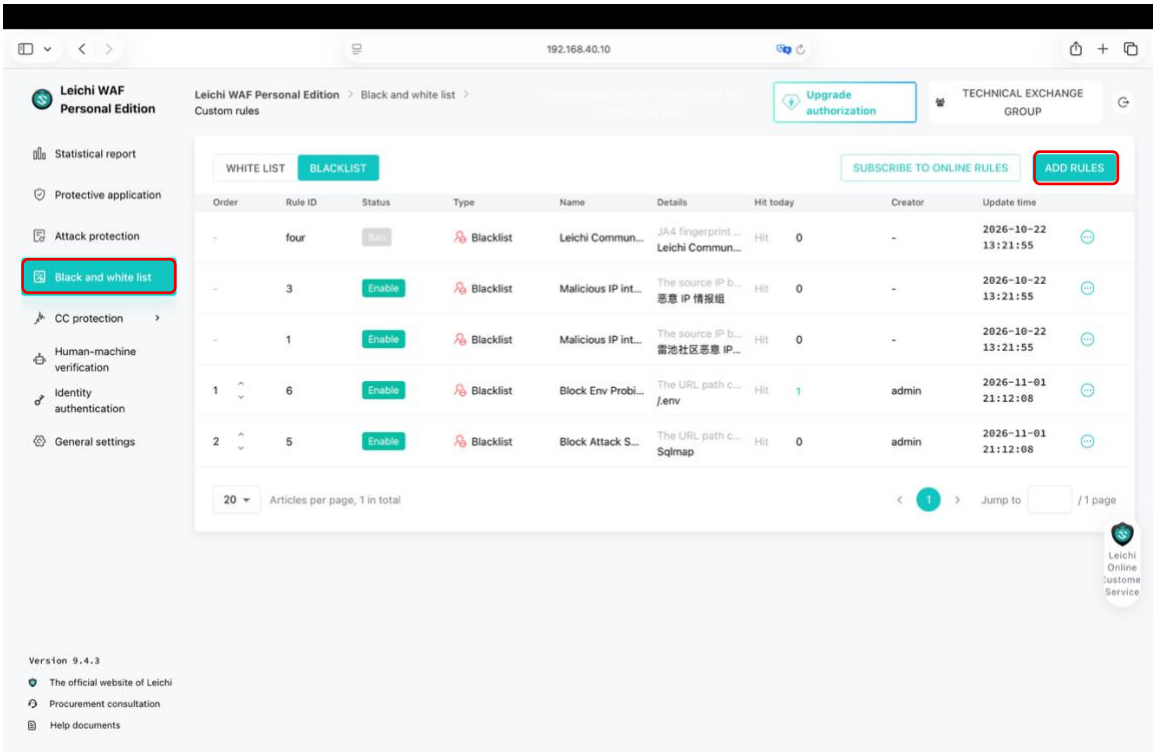
1. Step 1



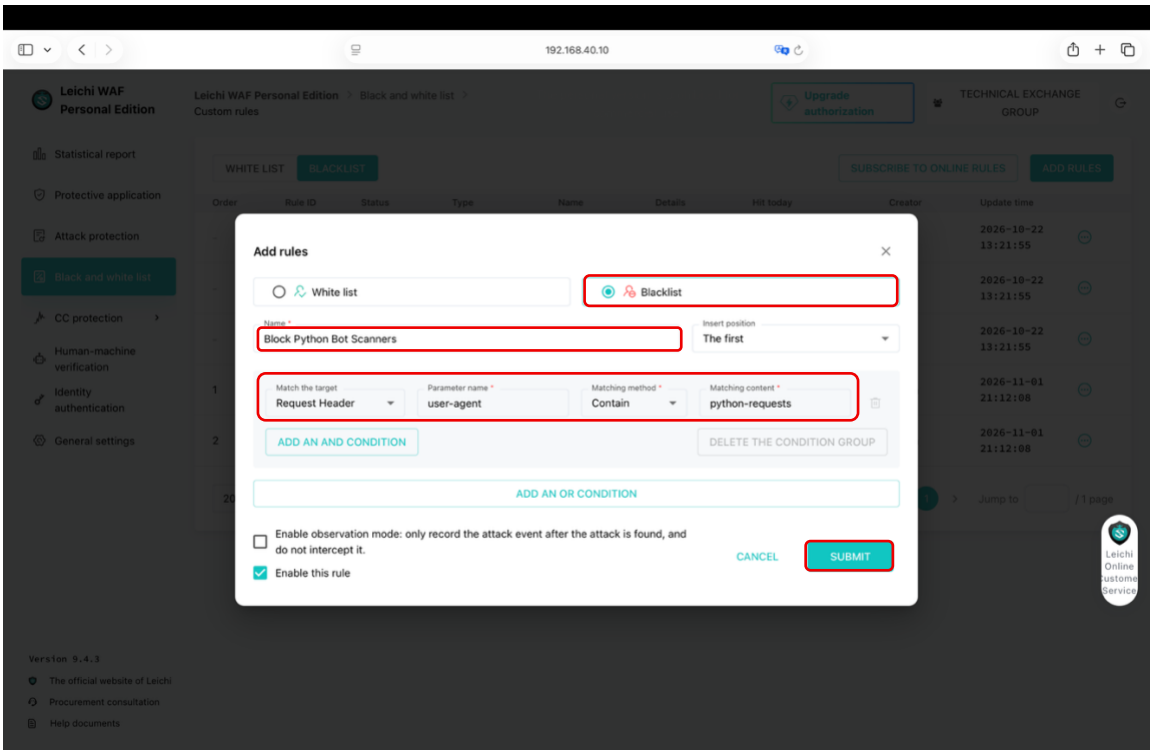
2. Step 2



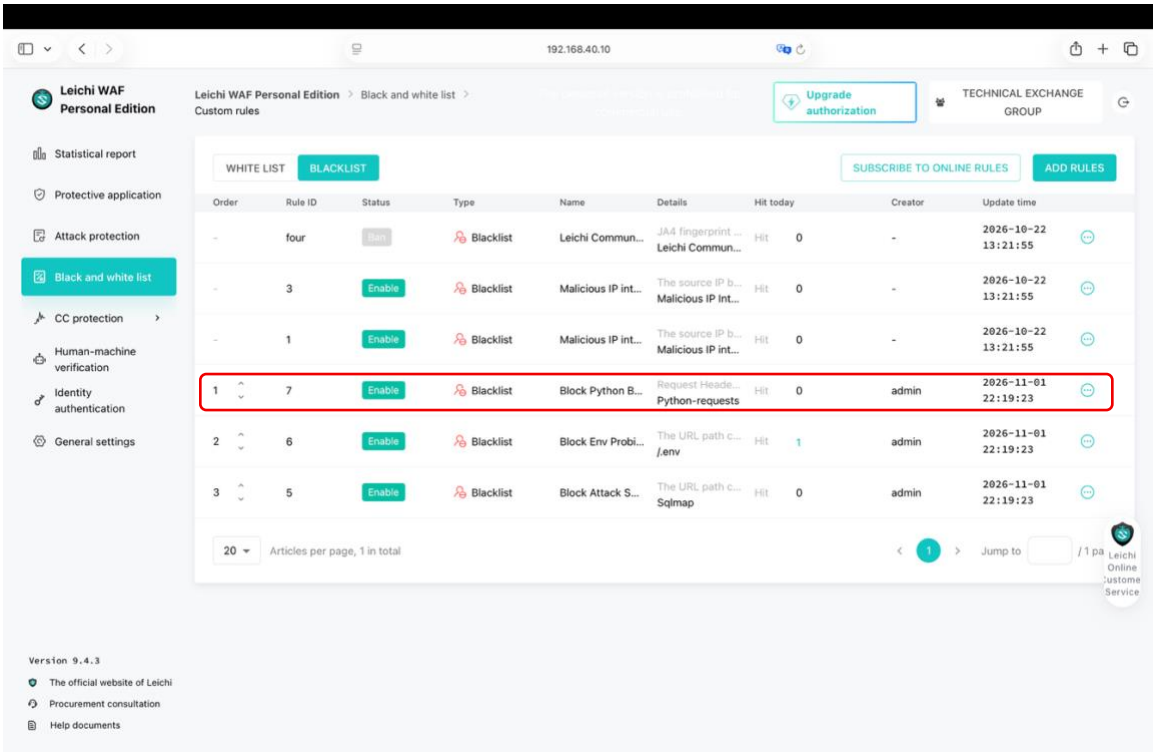
3. Step 3



4. Step 4

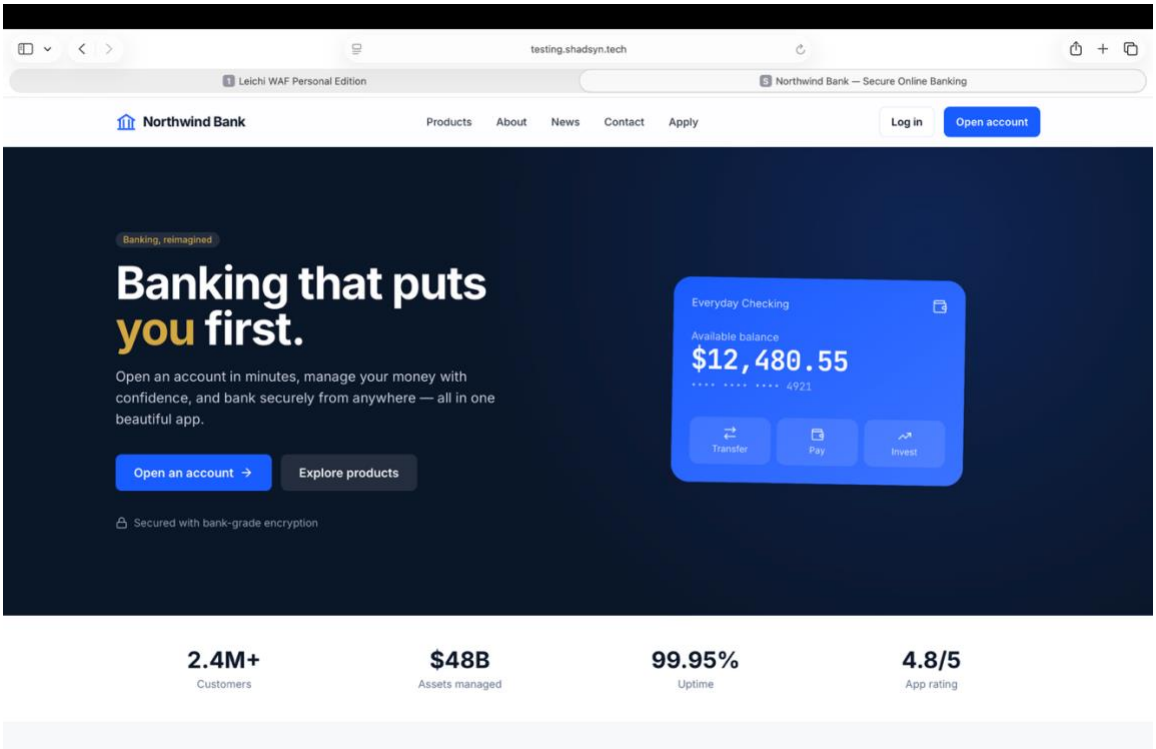


5. Step 5

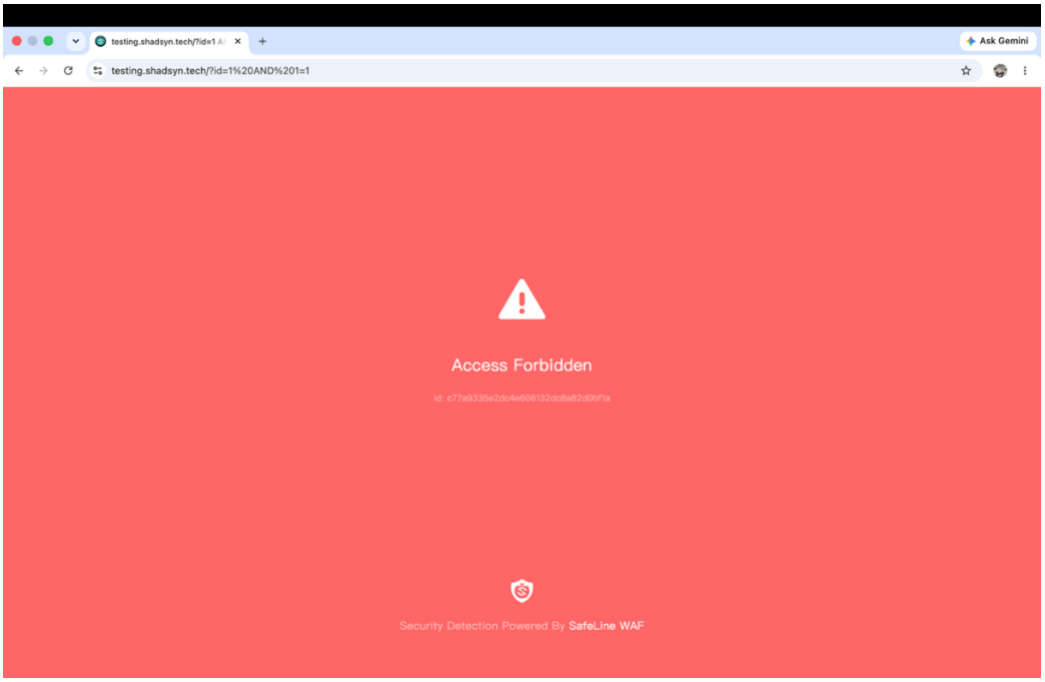


5. Testing

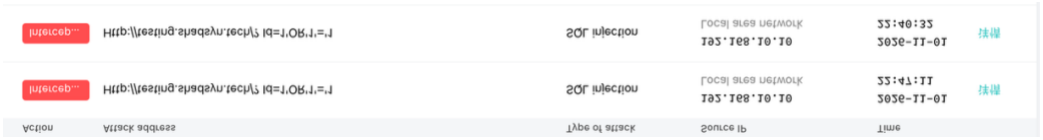
1. Normal



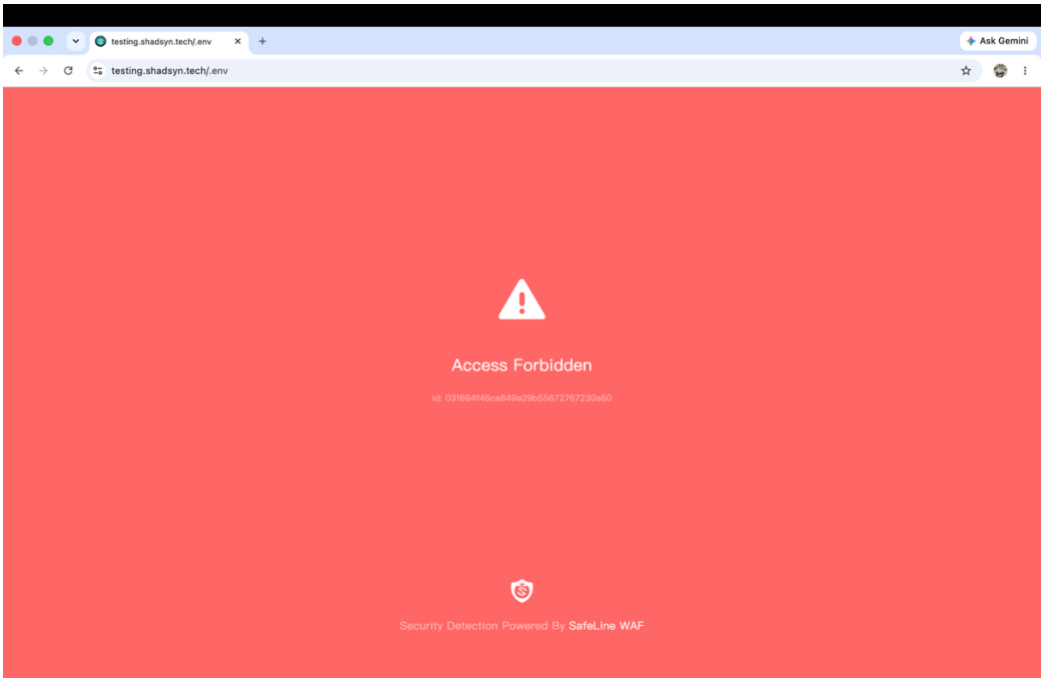
2. **Test block SQL Injection**



Attack Protection in SafeLine



3. **Test block .env**



Attack Protection in SafeLine

Action	Visit address	Type	Rule name	Source IP	Time	
Intercep...	Http://testing.shadsyn.tech/? Id=1'OR'1'='1	Blacklist	Block Python Bot Scanners	192.168.10.10 Local area network	2026-11-01 22:46:39	Details
Intercep...	Http://testing.shadsyn.tech/robots.txt	Blacklist	Block Python Bot Scanners	192.168.10.10 Local area network	2026-11-01 22:46:39	Details
Intercep...	Http://testing.shadsyn.tech/	Blacklist	Block Python Bot Scanners	192.168.10.10 Local area network	2026-11-01 22:46:38	Details

4. Test block Python bot Scanners



Attack Protection in SafeLine

Action	Visit address	Type	Rule name	Source IP	Time	
Intercep...	Http://testing.shadsyn.tech/? Id=1'OR'1'='1	Blacklist	Block Python Bot Scanners	192.168.10.10 Local area network	2026-11-01 22:46:39	Details
Intercep...	Http://testing.shadsyn.tech/robots.txt	Blacklist	Block Python Bot Scanners	192.168.10.10 Local area network	2026-11-01 22:46:39	Details
Intercep...	Http://testing.shadsyn.tech/	Blacklist	Block Python Bot Scanners	192.168.10.10 Local area network	2026-11-01 22:46:38	Details